OPERATION PRINCIPLES OF BI-DIRECTIONAL FULL-BRIDGE DC/DC CONVERTER WITH UNIFIED SOFT-SWITCHING SCHEME AND SOFT-STARTING CAPABILITY

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***Abstract---* A new bi-directional dual full-bridge dc/dc converter with a unified soft-switching scheme and soft-start capability is proposed in this paper. A simple voltage clamp branch is used to limit transient voltage across the current-fed bridge and realize zero-voltage-switching (ZVS) in boost mode operation, while achieve hybrid zero-voltage/zero-current switching (ZVZCS) for the voltage-fed bridge in buck mode operation. The theory of operations including soft-start-up process is discussed in this paper while the PWM control design and experimental results on 1.6 kW prototype will be presented in a companion paper.**

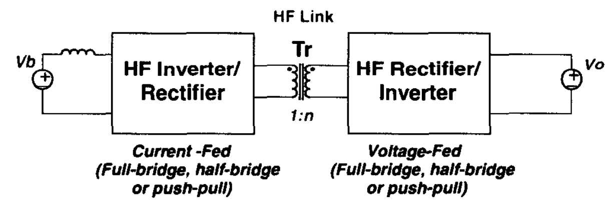
I. INTRODUCTION

A. High Power Bi-directional dc / dc Converter

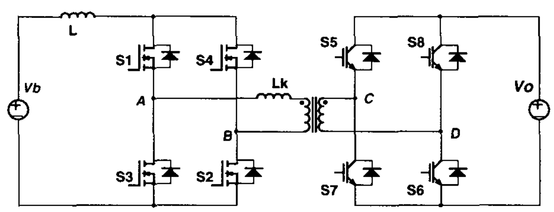
High power bi-directional dc/dc converters are needed for applications such as battery chargers/dischargers, uninterruptible power systems (UPSs), alternative energy systems, and hybrid electric vehicles. The fledgling nature of many of these application fields may be the main contributing factor for that only limited results are available on this aspect so far.

Most of the existing high power bi-directional dc/dc converters fall into the generic circuit topology illustrated in Fig. 1(a), which is characterized by a current-fed high-frequency (HF) inverter/rectifier on one side, preferably the lower voltage side, of the HF isolation transformer T_r, and a voltage-fed HF rectifier/inverter subtopology on the other side [17]. Each of these subtopologies can be either a full-bridge, a half-bridge or a push-pull circuit, or their variations. The current-fed half-bridge circuit is sometimes also referred as L-type boost circuit, or current-doubler. As an example, the bi-directional dc/dc converter with a full-bridge on each side and suitable for high power applications is shown in Fig. 1(b) [12].

The dual active bridge dc-to-dc converter proposed in [8-10], which has a voltage-fed bridge on each side of the isolation transformer seems an exception to the generic high power bi-directional dc/dc topology in Fig. 1(a). The operation of that circuit involves the utilization of the leakage inductance of the transformer as the main energy storing and transferring element, which puts severe burdens on the manufacturability and performance of the transformer. Moreover, power transferring is controlled by phase-shifting the voltages exerted on the two sides of the transformer, leading to the necessity to activate both bridges for power flow in either direction, offsetting the benefits of zero-voltage switching (ZVS), and also resulting in high energy circulation loss.



(a) Generic high power bi-directional dc / dc converter.



(b) Bi-directional full-bridge dc / dc converter
Fig. 1. High-power bi-directional dc / dc converters.

B. Isolated Boost (Current-Fed) Converter

In the generic bi-directional dc/dc converter discussed above, isolated boost or current-fed operation is inevitable. As is well known, isolated boost converter presents some severe performance limitations, i.e. high transient voltage and lack of self-starting capability. Because of the leakage inductance of the transformer, transient voltage happens on the switches on the current-fed side

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whenever the circuit tries to impress the boost inductor current through the transformer. Therefore, some form of voltage clamp circuit has to be incorporated to limit the transient voltage and ensure secure operation. A RCD clamp or snubber is the simplest way to solve the problem, but is not effective in high power operation. An autonomous "baby" buck converter was proposed to collect the transient energy and feed back to the source V_b [14]. Its control is self contained to limit the clamped voltage to some level and is independent of that for the main converter.

Soft-switching was thought to be the most effective way to solve the transient voltage problem. So far, only very few soft-switching schemes for isolated full-bridge boost converters have been successfully developed. The schemes proposed in [11]-[13] alleviate the transient voltage problem by either shortening or reverse the polarity of the secondary or the voltage-fed side of the transformer shortly before the turn-off of the current-fed side switches. These schemes need a bi-directional switch on the secondary side, which is readily available for the bi-directional dc/dc converter case. However, the voltage-side switches, normally high-voltage slower switches in high power applications, have to experience extra activation and hard turn-off, leading to lower converter efficiency. Moreover, some kind of voltage clamp mechanism is still necessary on the current-fed side.

The schemes presented in [6-7] and [18] employ an active clamp branch across the current-fed side switches and the transient voltage is clamped elegantly. Besides, the current-fed side switches are all turned on under ZVS. However, no PWM scheme with soft start-up capability is given [17].

C. Voltage-Fed Full-Bridge Converter

In contrast to the isolated current-fed converter, numerous ZVS voltage-fed (buck) full-bridge dc/dc converters with the well-know phase-shift PWM control have been developed in the past one and half decades, pioneered by the works in [1-2]. For high power applications where minority carrier devices such as IGBTs are frequently used, it has been shown that the combination of ZVS and zero-current-switching (ZCS), or called ZVZCS, is even more effective to reduce both the conduction loss and turn-off loss in the voltage-fed full-bridge converters. Some typical techniques to achieve ZVZCS are found in [3-5]. Because these soft-switching schemes for the voltage-fed converters are more familiar, they will not be reviewed here.

D. Contributions of the Paper

The proposed bi-directional full-bridge dc/dc converter in this paper, shown in Fig. 2, incorporates a unified soft-switching scheme. A simple clamp branch is used to achieve soft-switching operation in both directions of power flow. In buck mode of operation, the converter achieves hybrid zero-voltage and zero-current switching (ZVZCS) for the voltage-fed side switches by using a control timing modified from what was proposed in [4]; synchronous rectification can also be used for the current-fed side switches to reduce the conduction loss if they are implemented with MOSFETs. In boost mode of operation, the same clamp branch is mainly used to limit the transient voltage on the current-fed side switches which is intrinsic to the isolated boost type of converters; at the same time, zero-voltage switching (ZVS) is also realized for all the current-fed side switches, including the auxiliary clamp switch [6-7] [18]. A new non-phase-shifting PWM is used to seamlessly control the boost mode in both start-up and regular boost operation. Because soft-switching and voltage clamping are achieved in both directions of power flow, no lossy snubbering component is necessary. Reliable operation and high efficiency of the proposed converter are verified on a prototype designed for alternative energy applications. In the rest of the paper, the operation principles in each direction will be addressed, while the implementation and experimental results will be presented in a companion paper [15].

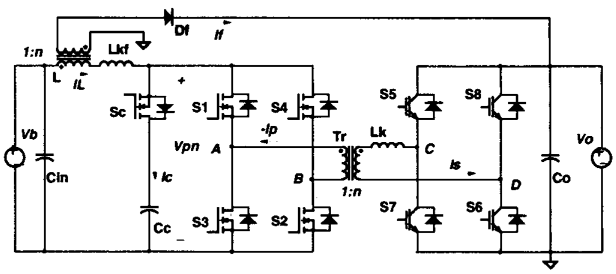

Fig. 2. Bi-directional full-bridge dc/dc converter with a unified soft-switching scheme and soft-start capability.

II. BUCK MODE OPERATION WITH ZVZCS AND SYNCHRONOUS RECTIFICATION

As shown in the circuit schematic in Fig. 2, the converter consists of two active bridges placed on each side of the main transformer T_r. Although not necessary, it is usually preferred to place the voltage-fed bridge on the higher voltage side, while the current-fed bridge on the current-fed side. The main reasons for this are easier smoothing and filtering of high current on the lower voltage side, lower switching loss because lower voltage rating switches can be used for the low voltage side switches, and better switch availability for the high side switches. In the implementation example shown in Fig. 2, high (voltage-fed) side switches are implemented with IGBTs and low (current-fed) side switches are implemented with MOSFETs. The flyback start-up winding on the main choke L is only used in the start-up stage in boost mode operation, while remains dormant in other operation modes. In the following discussions, all the active devices are assumed to be ideal switches. The transformer leakage inductance is reflected to one side of the transformers, and is illustrated in Fig. 2 as L_k and L_{kf} respectively.

In the buck mode of operation, the auxiliary (clamp) switch S_c is only activated briefly after the execution of an active duty-ratio to reset the transformer leakage current which otherwise circulates in the voltage-fed side bridge during the off duty-cycle, and at the same time, realize zero-current-switching (ZCS) for one pair of the switches, i.e. the so-called lagging leg switches, S_8 and S_6 in Fig. 2.

It should be mentioned that the concept to achieve ZVZCS with the help of an active clamp branch on the current-fed side was first introduced in [4]. In the scheme presented below, the timing for the clamp switch is modified such that it is aligned with the turn-off of the leading leg switches instead of in alignment with the turn-on of the lagging leg switches in [4]. As a result, an extra commutation between the rectifiers and the clamp switch on the current-fed side and the corresponding reverse-recovery loss are eliminated. Moreover, the voltage across the current-fed bridge is also made smother and cleaner.

A. ZVZCS Operation without Synchronous Rectification

The timing diagram and key circuit waveforms are shown in Fig. 3. Operation without synchronous rectification, i.e. without activation of S_1 through S_4, will be discussed first.

[t_0 - t_1]: Before t_0, S_5 has already been on, while the load current is freewheeling on the current-fed bridge. At t_0, S_6 is turned on, and the voltage V_o is immediately exerted on the transformer. Because the other side of the transformer is still shorted, the whole voltage is exerted on the leakage inductance L_k and causes the current to rise with the slope decided by V_o/L_k. With the transformer current increased linearly to the load current level at t_1, the freewheeling current shunts away from the pair of diagonal switches consisting of S_3 and S_4, and voltage across the transformer terminals on the current-fed side changes immediately to reflected voltage from the voltage-fed side, i.e. V_o/n, where n is the turns ratio of the transformer.

[t_1 - t_2]: Once the transformer current reaches the load current level at t_1, and V_{pn} rise to the reflected activating voltage V_o/n, the anti-parallel diode of the clamp switch starts to conduct the resonant current between L_k and the clamp capacitor C_c, assuming the main choke L is much higher in magnitude than the leakage inductance of the transformer. This process ends at t_2 when the resonation goes through a half resonant cycle and is blocked by the clamp switch S_c.

[t_2 - t_3]: After t_2, the circuit continues to run the normal active duty cycle as in the well known phase-shift controlled PWM ZVS full-bridge converter, and the choke continues to get charged.

[t_3 - t_4]: At t_3, the active duty-cycle ends with the turning off of S_5, and the potential of node *C* swings from the upper rail to the negative rail with the load current charging and discharging the parasitic capacitance of S_5 and S_7 respectively. The anti-parallel diode of S_7 then starts to conduct the freewheeling leakage current. Meanwhile, S_c is also gated on, and the voltage across the current-fed bridge, V_{pn}, which otherwise collapses down to zero, is held high. It is this voltage which is reflected to the transformer terminals on the voltage-fed side, *C* and *D*, and exerted on the leakage inductance and makes its current decrease.

[t_4 - t_5]: With the anti-parallel diode of S_7 conducting, S_7 can be turned on at t_4 (with a dead time relative to the turn-off of S_5) with ZVS. After t_4, the leakage current continues to reduce faster with the same rate as it rises at the beginning of the active duty cycle. This process proceeds until it is completely reset to zero and the anti-parallel diodes of S_1 and S_2 starts to reverse block and prevent leakage current to go into the other direction. During this time, the load current is increasingly provided by the clamp branch. As a result, the effective duty cycle seen on the current fed side is slightly larger than what is commanded by the PWM controller on the voltage-fed side.

[t_5 - t_7]: At t_5, S_c is turned off, and V_{pn} drops immediately to zero and the freewheeling duty cycle is initiated. This stage of operation is the same as in

regular phase-shift controlled ZVS full-bridge converters. During this time, there is no current freewheeling in the voltage-fed side bridge, saving the extra conduction loss.

$[t_7 - t_8]$: At t_7, S_6 is turned off with complete ZCS because the leakage current has already been reset before t_5. So IGBTs with lower current rating can be comfortably used for S_8 and S_6.

$[t_8 - t_9]$: At t_8, S_8 is turned on and the circuit assumes the exact same operation experienced in the previous half cycle. The only differences are that the active switches change to the other pair of diagonal switches, S_7 and S_8, S_3 and S_4, and the voltage on the transformer reverses polarity to exercise flux balance.

To summarize, the benefits of the discussed ZVZCS scheme are twofold. First, the turn-off loss in the converter is reduced by shifting half of the turn-off loss on the voltage-fed side, usually implemented with IGBTs, to the current-fed side, or more precisely, to S_c, which is usually implemented with fast switches such as MOSFETs. Second, the conduction loss during the off duty cycle on the voltage-fed side is completely eliminated. Besides, ZVS for the leading leg switches is still retained.

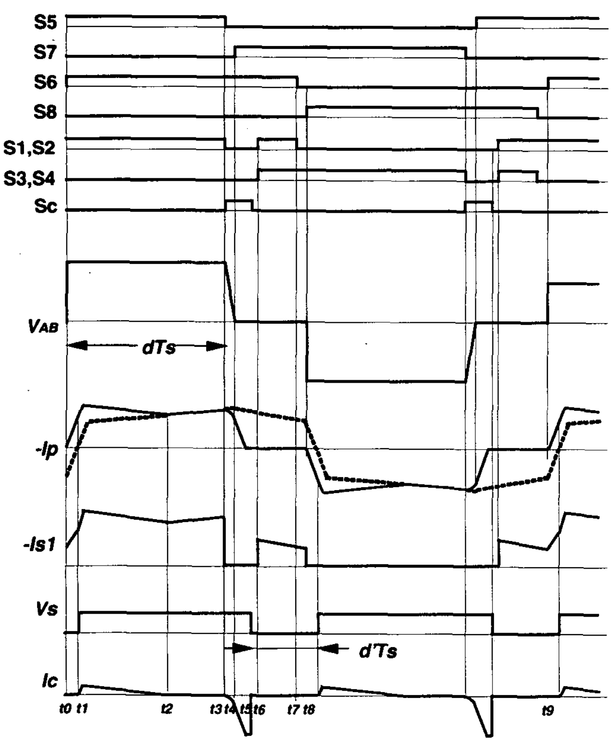

Fig. 3. Timing diagram of buck mode operation - ZVZCS and synchronous rectification.

B. ZVZCS Operation with Synchronous Rectification

If the current-fed side bridge switches are implemented with MOSFETs, their reverse conducting capability can be used together with their intrinsic diode to reduce the conduction loss in the bridge. The benefits of doing the so-called synchronous rectification can easily outweigh the extra driving loss in the interested operation frequency range.

The timing diagrams for S_1 and S_2, and S_3 and S_4 to achieve synchronous rectification are shown also in Fig. 3. As can be seen, S_1 and S_2 are simultaneously activated during the positive duty cycle and both off duty cycles except when the clamp switch is turned on. This exception is important to prevent the shoot-through of the current-fed bridge switches. The logic expressions to realize the above functions are

$$S_1, S_2 = (S_5 + S_6) \cdot \overline{S}_c, \tag{1}$$

and

$$S_3, S_4 = (S_7 + S_8) \cdot \overline{S}_c. \tag{2}$$

In reality, a short dead time between the end of S_c and the onset of S_1 through S_4 is needed to prevent shoot-through. The current flowing through the MOSFETs is also illustrated in Fig. 3 where it is assumed that when the MOSFETs are on, all the load current goes through the MOSFETs instead of their body diode.

III. BOOST MODE OPERATION WITH ACTIVE VOLTAGE CLAMP

The circuit operates as an isolated boost full-bridge converter when power is transferred from the current-fed bridge side to the voltage-fed bridge side. All the isolated boost converters (full-bridge, push-pull and L-type) inevitably suffer from high transient voltage, and voltage clamping schemes need to be incorporated to limit that voltage across the current-fed bridge switches. One of the effective schemes is to use an active clamp branch across the bridge, which consists of an active switch in series with a capacitive energy storage component, S_c and Cc as shown in Fig. 2.

Another special problem with the isolated boost converter is the start-up problem before the output voltage is established. That is, the boost choke current needs to be properly reset. One solution to the problem is shown in Fig. 2. An extra flyback winding is added on the main choke, and it can be connected either to the supply side, or to the output through a rectifying diode. If it is connected to the supply side, the supply current becomes continuous during start-up, facilitating input filtering. However, the choke energy is circulating in the circuit. By connecting that winding to the output, as shown in Fig. 2, the choke current is reset by the rising output voltage, and its energy is fed to the output in the start-up process, accelerating the establishment of the output voltage. The only shortcoming with this scheme is that the input (choke) current becomes discontinuous, or bi-directional if active clamp branch is used, and the input filtering requirement is higher. But the start-up process is usually very brief, and filter loss should not be a concern.

The same clamp branch is activated during both start-up and regular boost operation modes; however, as discussed later, its position relative to the main bridge switches changes. The voltage-fed bridge switches are either disabled or used simply as rectifiers, or can be activated to achieve synchronous rectification if implemented with MOSFETs.

A (non-phase-shift) PWM control is developed to control the circuit to achieve smooth transition from start-up to regular boost operation mode, ZVS for all the active switches, and minimal conduction loss.

A. Start-up Operation with Active Clamp

The timing diagram and key circuit waveforms during start-up operation are shown in Fig. 4. The operation principle is very similar to the active clamp flyback converter, except that in the on duty cycle, energy is also transferred to the output through the main transformer. As a result, its behavior and transfer characteristics are the same as a Weinberg converter with an active clamp. As shown below, the whole voltage transfer characteristics are equivalent to a buck or forward converter. The operation principles are explained below with reference to Figs. 1 and 3. It is assumed that the circuit is running under continuous current mode, i.e. the magnetizing current or the current flowing through the secondary flyback winding will not decay to zero during the off duty cycle.

[t_0 - t_1]: Before t_0, S_c was on, while all the bridge switches were off. Load current flows through the flyback winding to the output, while the clamp capacitor current, I_c, and the choke leakage current, I_L, were equal and negative, and the energy stored in the clamp capacitor was fed back to the supply V_b. At t_0, S_c is turned off, and I_c immediately goes to zero. The negative current of the leakage inductance of the choke resonates with the parasitic capacitance (not shown in the schematics) across p and n, which brings V_{pn} down to zero at t_1.

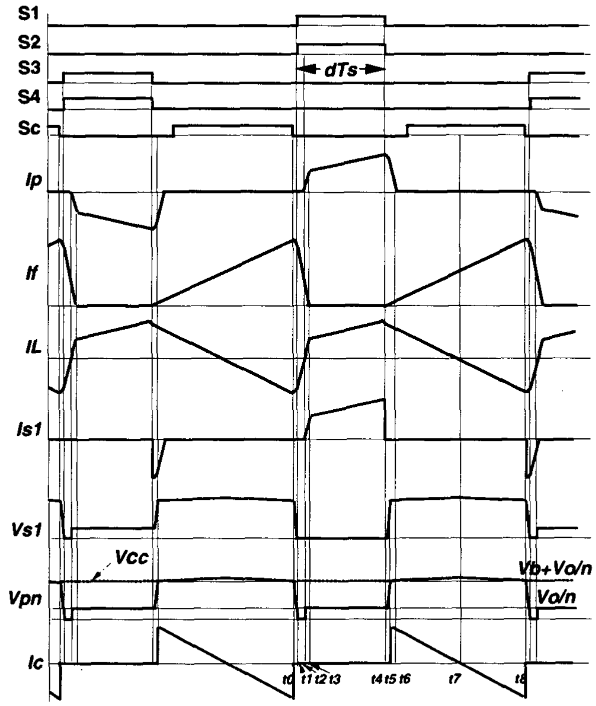

Fig. 4. Timing diagram of start-up operation in boost mode.

[t_1 - t_2]: Assuming that the leakage energy in L_{kf} is larger than the capacitive energy stored in the stray capacitance across p and n, I_L is still negative after V_{pn} is reduced to zero. The activating voltage on L_{kf} will be $V_b + V_o/n$, and makes it increase linearly until t_2 when it increases to zero. During this interval, S_1 and S_2 can be turned on simultaneously under zero-voltage.

[t_2 - t_3]: After t_3, S_1 and S_2 start to carry the choke current which is linearly increasing, while the primary of the main transformer T_r is clamped to the reflected output voltage, V_o/n. At t_3, the leakage current of the choke is equal to its magnetizing current and the flyback winding current is reduced to zero.

[t_3 - t_4]: The circuit behaves similar to the on duty cycle of a boost converter, but the choke is charged with the difference between the supply voltage and the reflected output voltage, i.e. V_b - V_o/n.

[t_4 - t_5]: At t_4, S_1 and S_2 are turned off, and V_{pn} is charged linearly to $V_b + V_o/n$, and further clamped by the clamp capacitor at t_5.

[t5 - t6]: Starting at t_5, the choke leakage inductance L_{kf} resonates with the clamp capacitor C_c, and its current starts to get reset. The difference between the magnetizing current of the choke and the leakage current flows through the secondary flyback winding. Independently, the sum of the reflected clamp voltage and the output voltage quickly resets the main transformer leakage current.

[t_6 - t_8]: The clamp switch S_c can be turned on at anytime between t_5 and t_7 with ZVS. With the resonance between L_{kf} and C_c continuing, L_{kf} is reset to zero at t_7 and goes negative. The clamp voltage is usually not modified considerably by the resonance because the energy stored in C_c is usually much larger than the leakage energy. At t_8 the next half cycle is initiated with the turn-off of S_c.

In reality, the time intervals from t_0 to t_3 and t_4 to t_6 are very brief compared with the rest of the PWM cycle. Because of the fast commutations, severe voltage ringing may still happen at t_5 when the choke magnetizing and leakage energy is dumped into the clamp capacitor through the antiparallel diode of S_c. For that reason, layout efforts are needed to minimize the stray inductance in the loop formed by input filter capacitor C_{in}, choke L, and the clamp branch. Also, during the on-time of the bridge switches, e.g. between t_1 and t_4, the main choke is in series with the leakage inductance L_k of the main transformer while the clmap branch is disengaged. The stray capacitance across p and n, and the leakage inductance L_k form a resonant tank, and make the transformer and voltage V_{pn} oscillatory.

The dc transfer characteristics of the circuit during start-up mode can be easily derived using the flux balance condition on the boost choke. Assuming the circuit runs in continuous current mode and neglecting the effects of the leakage inductance L_{kf}, during the on-time of the bridge switches, defined as dT_s in Fig. 4, the flux increase on the choke is

$$\Delta\phi_1 = (V_b - V_o / n)dT_s, \tag{3}$$

while during the off-time, $(1/2-d)T_s$, the flux decrease is

$$\Delta\phi_2 = (V_o / n)(0.5 - d)T_s, \tag{4}$$

where n is the turns ratio of the transformer defined in Fig. 2, and T_s is the PWM period. Solving the above equations by letting $\Delta\phi_1 = \Delta\phi_2$,

$$V_o = 2d(nV_b), \tag{5}$$

where $0 < d < 0.5$. It turns out that the dc transfer function is simply that of a buck or forward converter with $2d$ being its effective duty cycle (ratio). After d reaches 0.5, the circuit can run in its regular boost mode.

B. Boost Mode Operation with Active Clamp

When the output voltage is higher than nV_b, the flyback winding path is automatically disengaged from the circuit, and the circuit runs in regular isolated boost converter mode. During boost mode operation, the bridge switch pairs in diagonal positions operating at duty cycles larger than 0.5. The boost choke is charged during the overlapping period when all the four bridge switches are on, and discharged when one diagonal switch pair is switched off and the clamp switch is turned on. The timing diagram and circuit waveforms are shown in Fig. 5.

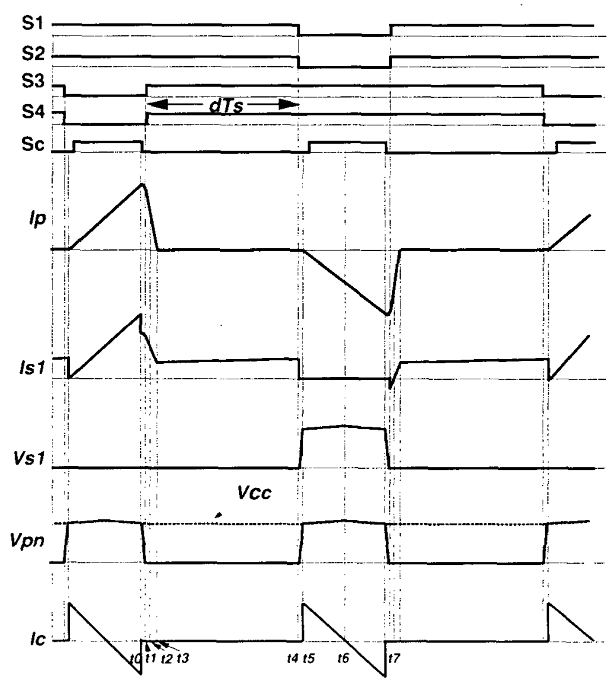

Fig. 5. Timing diagram of boost mode operation.

[t_0 - t_1]: Before t_0, clamp switch S_c and bridge switches S_1 and S_2 were on. The energy stored in the choke and the clamp capacitor was discharged to the output and the current in the transformer winding was higher than the choke current, with the difference supplied by the clamp branch. At t_0, S_c is switched off, and suddenly makes the transformer leakage current higher than the choke current. The surplus current discharges the parasitic capacitance across p and n, and the bridge switches S_3 and S_4,

resulting in V_{pn} resonating down to zero at t_1. Again assuming the leakage energy is larger than the stray capacitance energy, the leakage current is still higher than the choke current at t_1.

$[t_1 - t_2]$: After t_1, the transformer current remains reset, and it equals the choke current at t_2. It is during this interval that S_3 and S_4 can be turned on with ZVS because their antiparallel diodes are conducting.

$[t_2 - t_4]$: The transformer leakage current is reset to zero at t_3. With all the bridge switches on, the choke is charged during this interval in the same way as in regular boost converter.

$[t_4 - t_5]$: At t_4, S_1 and S_2 are turned off, and the choke current charges V_{pn} up until it is clamped by the clamp capacitor at t_5.

$[t_5 - t_7]$: After t_5, the voltage difference between the clamp capacitor and the reflected output voltage is exerted on the leakage inductance of the transformer. If the voltage ripple on the clamp capacitor is neglected, the transformer current will ramp up linearly. The clamp branch current reverses its direction at t_6, and the transformer current is higher than the choke current.

At t_7, the clamp switch is turned off, and it initiates the next half cycle of operation.

C. Clamp Branch

As is clear from the above analysis, the clamp switch S_c is turned on under ZVS while turned off under hard switching, so MOSFETs with the same voltage rating as the main bridge switches can be adequately used. If turned on earlier during the active (on) duty time in the regular boost mode, and during the freewheeling (off) stage in the start-up mode, it can also be used to do synchronous rectification. If in some applications where high voltage MOSFETs (usually with slow recovery intrinsic body diode) have to be used, external low voltage Schottky diodes can be placed in series with the MOSFETs to block the internal diodes of the MOSFETs and external fast recovery diodes are used instead to prevent potential failure during dynamic transients.

The clamp capacitor C_c can be selected according to the requirements for operation in boost mode. Usually, the leakage inductance, L_{kf}, of the coupled choke for start-up is much larger than that of the transformer, L_k, because gapped cores have to be used to achieve the required magnetizing inductance, so design can be based solely on the resonant tank formed by C_c and L_k. The resonance happens during the off stage of boost mode operation, i.e. between t_5 and t_7 in Fig. 5, with its maximum span being about $T_s/2$ when the circuit just turns into regular boost mode from start-up mode. The criterion is to select C_c such that the resonant period is larger than $T_s/2$, or

$$C_C \geq (T_S / 4\pi)^2 / Lkp, \tag{6}$$

where $L_{kp} = L_k/n^2$ is the transformer leakage inductance reflected to the current-fed side, and T_s is the period of the driving signal for each bridge switch.

The timing signal of S_c for each of the operating modes is easy to generate. However, its position relative to the main bridge switches involves a swapping when the duty cycle reaches the boundary where the circuit changes from start-up to regular boost mode. As a result, extra logic is needed to ensure the smooth transition. The issue will be addressed in further detail in the companion paper.

IV. SUMMARY

A new bi-directional dual full-bridge dc/dc converter with a unified soft-switching scheme and soft-start capability is proposed in this paper. The bridge on one side, preferably the lower voltage side, is current-fed, while that on the other side is voltage-fed. A simple voltage clamp branch, which is composed of an active switch with its antiparallel diode and a capacitive energy storage element in series, is placed across the current-fed bridge to limit transient voltage across the current-fed bridge and realize zero-voltage-switching in boost mode operation, while achieve hybrid zero-voltage/zero-current switching (ZVZCS) for the voltage-fed bridge in buck mode operation. The detailed operation principles of the converter are discussed.

In buck mode operation, the voltage-fed bridge is controlled by the well-known phase-shift PWM. The clamp branch is activated only briefly each time after an on duty cycle is executed and the on-time of the clamp switch is just long enough to reset the transformer leakage current to zero and achieve ZVZCS operation even under maximum load current.

In boost mode operation, a non-phase-shift PWM control scheme together with an extra flyback winding coupled to the main boost inductor is developed to realize soft-starting and then naturally transfer into regular boost mode when the switch duty cycle reaches 0.5.

The implementation issues, PWM control and experimental results on a 1.6 kW prototype will be presented in a companion paper [15].

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